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Winter, 2025

B. Tech 2nd Semester Examination

Mathematics II (Theory)

Course Code: BMA23112T

Full Marks – 50

Time –2 hours

The figure in the margin indicates full marks for the questions.

SECTION A

1. Choose the correct Option:

1X10=10

(i) The order and degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 - \left(\frac{dy}{dx}\right)^4 = x^4$ are

- (a) 2, 1
- (b) 2, 2
- (c) 2, 3
- (d) 2, 4

(ii) The condition for the differential equation $P(x,y)dx + Q(x,y)dy = 0$ to be exact is

- (a) $\frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y}$
- (b) $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$
- (c) $\frac{\partial P}{\partial x} = \frac{\partial Q}{\partial x}$
- (d) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial y}$

(iii) The integrating factor of the homogeneous equation $M(x,y)dx + N(x,y)dy = 0$ is

- (a) $\frac{1}{My+Nx}$
- (b) $\frac{1}{Mx+Ny}$
- (c) $\frac{1}{My-Nx}$
- (d) $\frac{1}{Mx-Ny}$

(iv) The complementary function of the equation $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6 = x^2$ is

- (a) $C_1 \cos 2x + C_2 \sin 3x$
- (b) $C_1 e^{2x} + C_2 e^{3x}$
- (c) $C_1 e^{-2x} + C_2 e^{-3x}$
- (d) none of the above

(v) The value of the Bessel's function $J_1(0)$ is

- (a) 0
- (b) 1
- (c) -1
- (d) $(-1)^n$

(vi) The value of $\lim_{z \rightarrow i} \frac{z^2+1}{z-i}$ is

- (a) i
- (b) $-i$
- (c) $2i$
- (d) $-2i$

(vii) $\frac{d}{dz} (-z)^n =$

- (a) $(-1)^n z^n$
- (b) $(-1)^n z^{n-1}$
- (c) $(-1)^n n z^n$
- (d) $(-1)^n n z^{n-1}$

(viii) The Cauchy-Riemann equations of the analytic function $f(z) = u + iv$ are

- (a) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial x}, \frac{\partial u}{\partial y} = \frac{\partial v}{\partial y}$
- (b) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial x}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial y}$
- (c) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$
- (d) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

(ix) The function $f(z) = \frac{e^z}{(z+1)^4}$ has a pole

- (a) at $z = 0$ of order 4
- (b) at $z = -1$ of order 4
- (c) at $z = 1$ of order 4
- (d) none of the above

(x) The value of the integral $\oint_C \frac{e^z}{z-2} dz$ is where C is the circle $|z| = 1$ is

- (a) 0
- (b) e
- (c) $2\pi i$
- (d) $2\pi ie$

SECTION B

2. Answer any **one** question:

(a) (i) Solve the equation $\frac{dy}{dx} + 2y \tan x = \sin x$; given $y = 0$ when $x = \frac{\pi}{3}$

(ii) Solve the equation $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$

(b) (i) Solve the equation $(y+x)dx + xdy = 0$

(ii) Prove that the equation $(x^2 + y^2 + 2x)dx + 2ydy = 0$ is not exact. Find a suitable integrating factor and hence solve it.

(c) (i) Solve: $p^2 + p \left(\frac{y}{x} - \frac{x}{y} \right) = 1$; $p = \frac{dy}{dx}$

(ii) Solve: $p = \log(xp - y)$ OR $x = py - p^2$; $p = \frac{dy}{dx}$

SECTION C

3. Answer any **one** question:

- (a) (i) Find the complete solution of the equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = e^x$ 4
(ii) Solve: $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} = x$ 6
- (b) (i) Using method of variation of parameters solve the equation $\frac{d^2y}{dx^2} + 4y = \tan 2x$ 7
(ii) Prove that the Legendre's polynomial $P_2(x) = \frac{1}{2}(3x^2 - 1)$ 3
- (c) (i) Find the series solution of the equation $\frac{d^2y}{dx^2} + x^2y = 0$ 7
(ii) Prove that the Bessel's function $J_0(0) = 1$ 3

SECTION D

4. Answer any **one** question:

- (a) (i) Prove that the function $f(z) = \bar{z}$ is analytic nowhere. 4
(ii) Define a harmonic function. Prove that the function $\phi = x^2 - y^2$ is harmonic. 1+5=6
- (b) (i) Find the imaginary part v of the analytic function $f(z) = u + iv$ if $u = 2x(1 - y)$ 4
(ii) Find the analytic function $f(z) = u + iv$ where $u = \log(\sqrt{x^2 + y^2})$ 6
- (c) (i) Determine the region in w -plane bounded by the lines $x = 0, y = 0, x = 2, y = 1$ in z -plane under the transformation $w = z + (2 + 3i)$ 5
(ii) Find the image of the circle $|z - 2i| = 2$ under the transformation $w = u + iv = \frac{1}{z}$ 5

SECTION E

5. Answer any **one** question:

- (a) (i) Evaluate $\int_C \frac{z^2 dz}{z^2 - 4}$ where C is $|z - 2| = 1$ 5
(ii) Prove that $\frac{1+2z^2}{z^3+z^5} = \frac{1}{z^3} + \frac{1}{z} - z + z^3 - z^5 + \dots$ 5
- (b) (i) Evaluate $\int_C \frac{e^z}{z^2} dz$, where C is the positively oriented contour $|z| = 1$ 6
(ii) Determine the singularities of (i) $f(z) = \frac{e^z - 1}{e^z}$, (ii) $f(z) = \frac{e^z}{e^z - 1}$ 4
- (c) State Cauchy Residue theorem. Apply this theorem to evaluate the integral $\int_C \frac{2z-1}{z(z+1)(z-3)} dz$ where C is $|z| = 2$ 2+8=10
